

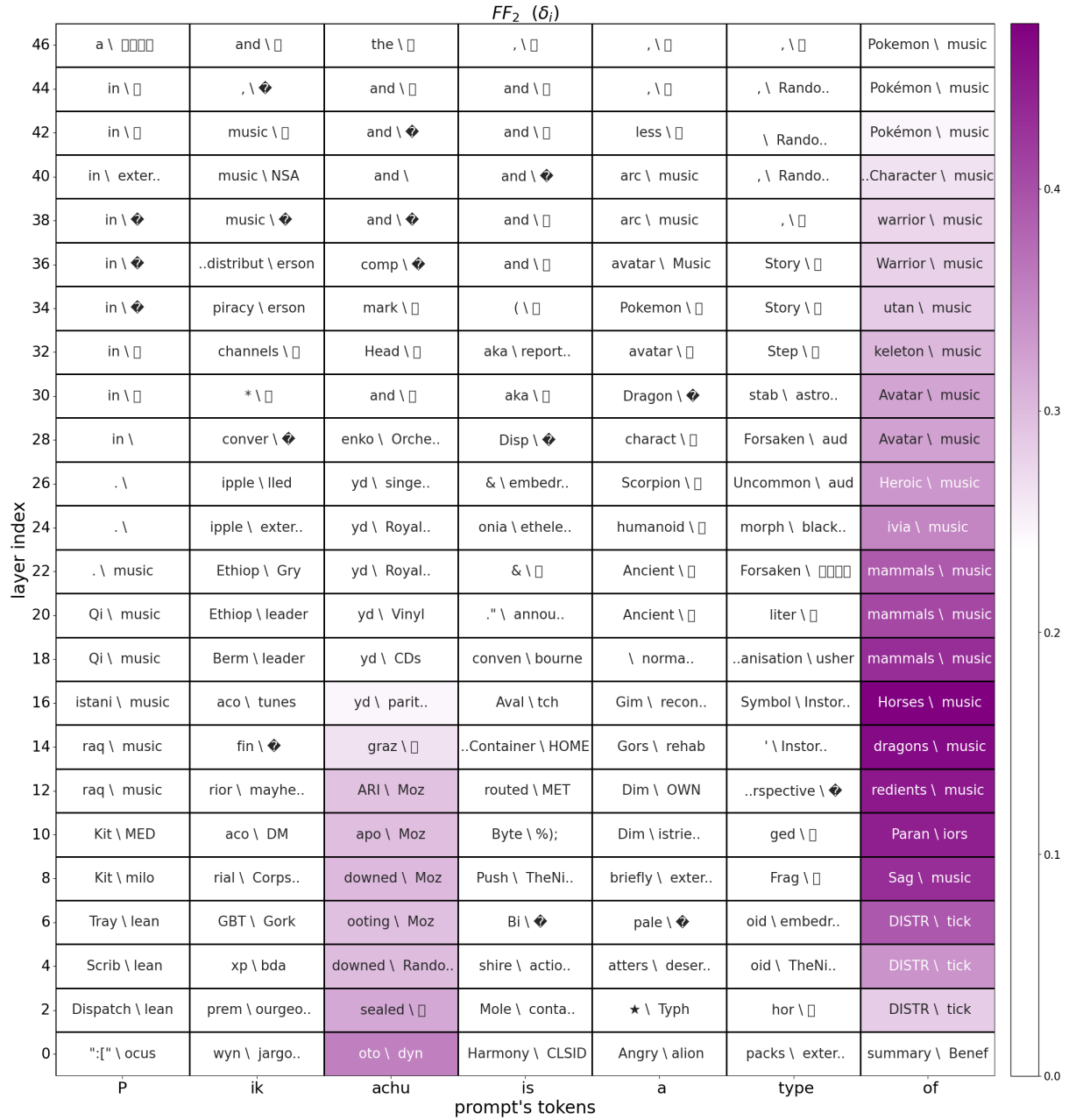


Figure 13: GPT2-xl FF_2 gradients via Logit Lens. The editing tries to change the prompt “Pikachu is a type of” to answer “music” instead of “Pokémon”. Every cell shows the most < probable \improbable > tokens for its δ_i . The fact that we see FF_2 ’s last token’s project “Pokémon / music” means the edit tries to subtract from the model’s weights the embedding of “Pokémon”, while adding the embedding of “music” (same mechanism with “mammals / music”). Regarding the presence of many non-English tokens (question marks and empty boxes), in [Appendix H](#), we present the same table with “Normalized Logit Lens”, revealing the embedding of the target token “music”.

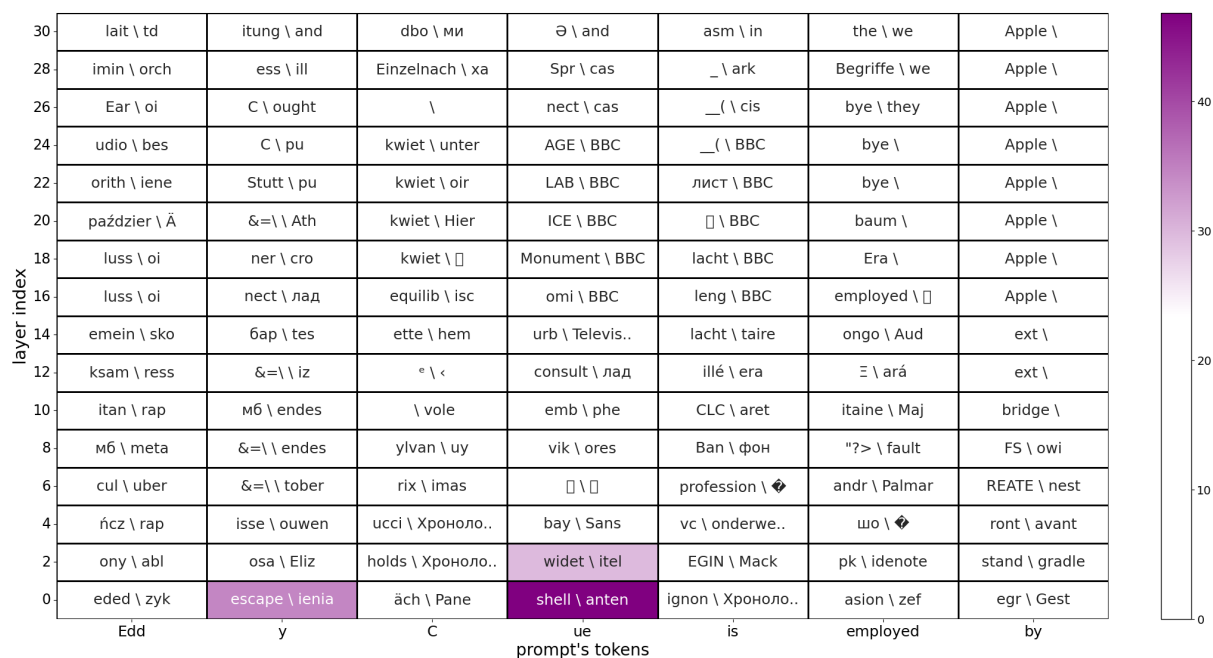


Figure 14: Llama2-7B FF_2 gradients via Logit Lens. The editing tries to change the prompt “Eddy Cue is employed by” to answer “BBC” instead of “Apple” (template was taken from the CounterFact dataset). Every cell shows the most < probable \ improbable > tokens for its δ_i . 3 main patterns can be observed from the table: (1) The part that has the largest VJPs by norm, which we assume is where most of the editing is done, is in the last subject token (“cu”) and during the first few layers of the model. (2) We do not see the target token “BBC” in the most improbable projection of the last token; instead, the most improbable one is the empty token “”. When we examined these projections we found “BBC” to be only the third most improbable token. However, in other VJPs we notice this is indeed the most improbable projection. (3) The original answer of the model, “Apple”, frequently emerges in the projections of the last token. As outlined in Section 5.2, given that we update the model with a negative learning rate, when the VJPs Logit Lens projection ranks a token as highly probable, subtracting the gradients from the model’s weights equals to subtracting the embedding of this probable token from them. Consequently, this pattern illustrates the mechanism of “shift” within the context of “imprint and shift”: We decrease the probability associated with “Apple” in pursuit of increasing the probability of “BBC”.

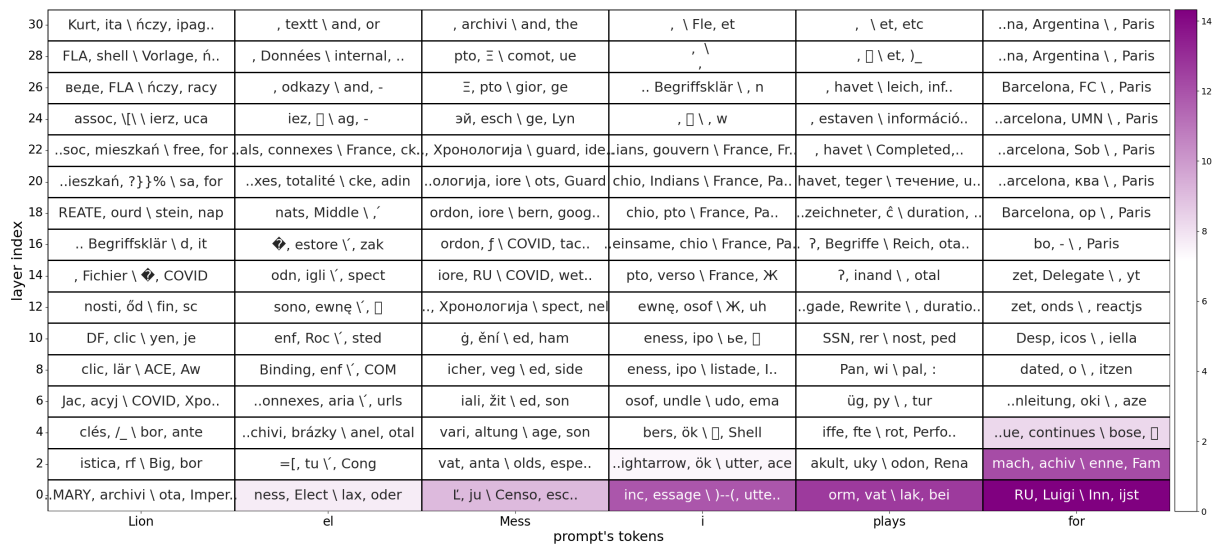


Figure 15: Llama2-7B FF_2 gradients via Logit Lens. The editing tries to change the prompt “Lionel Messi plays for” to answer “Paris” rather than “Barcelona”. Every cell shows the **2 most probable and improbable tokens** of the VJPs δ_i in the format of $\langle \text{probable} \backslash \text{improbable} \rangle$. The target token, “Paris” is the second most improbable token in the projections of the last prompt’s token (“for”), only second to the empty token (“”). The projection’s most probable tokens of the last prompt’s token are “Barcelona” (the final prediction of the model for this prompt), with some projections also revealing “Argentina” (which was the second most probable prediction of the model). In addition, the projections of the last subject token from the prompt (“i”) show that among the most improbable tokens there are appearances of “France”, which can be associated to its capital, “Paris”. Together, these projections illustrate how backpropogation editing tries to add into the model’s FF_2 ’s neurons (weights) the embedding of “Paris” (and related concepts, e.g. “France”) while removing the embeddings of “Barcelona” and “Argentina”.